

# Digital Logic Circuits (Spring 2026)

## Quiz #3

Name: \_\_\_\_\_

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1. A full-adder is characterized by b
  - a) two inputs and two outputs
  - b) three inputs and two outputs
  - c) two inputs and three outputs
  - d) two inputs and one output
  
2. To expand a 2-bit parallel adder to a 4-bit parallel adder, you must d
  - a) use two 2-bit adders with no interconnections
  - b) use two 2-bit adders and connect the sum outputs of one to the bit inputs of the other
  - c) use four 2-bit adders with no interconnections
  - d) use two 2-bit adders with the carry output of one connected to the carry input of the other
  
3. In general, a multiplexer has d
  - a) one data input, several data outputs, and selection inputs
  - b) one data input, one data output, and one selection input
  - c) several data inputs, several data outputs, and selection inputs
  - d) several data inputs, one data output, and selection inputs
  
4. Each of the eight full-adders in an 8-bit parallel ripple carry adder exhibits the following propagation delay:

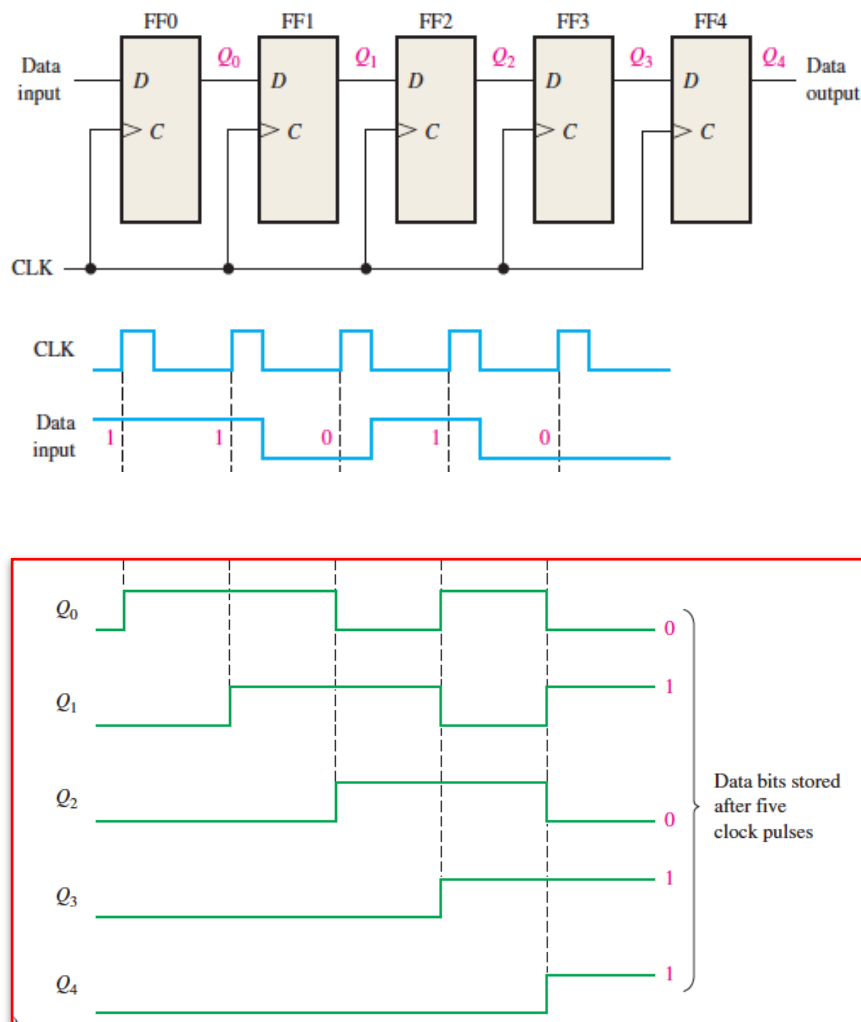
|                                 |       |
|---------------------------------|-------|
| $A$ to $\Sigma$ and $C_{out}$ : | 20 ns |
| $B$ to $\Sigma$ and $C_{out}$ : | 20 ns |
| $C_{in}$ to $\Sigma$ :          | 30 ns |
| $C_{in}$ to $C_{out}$ :         | 25 ns |

Determine the maximum total time for the addition of two 8-bit numbers.

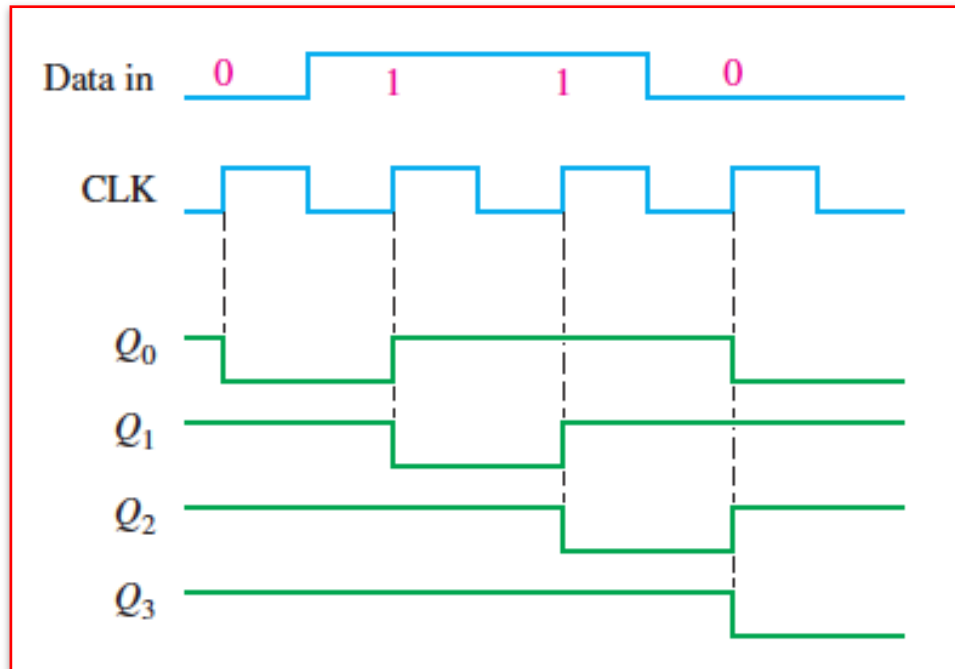
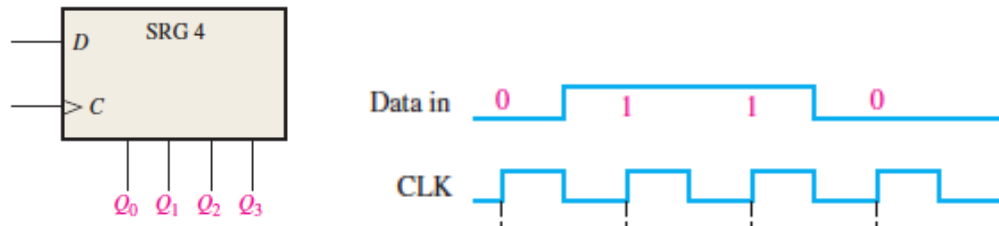
$$T_p = 20 \text{ ns} + 6(25 \text{ ns}) + 30 \text{ ns} = 200 \text{ ns}$$

5. The purpose of the clock input to a flip-flop is to d
  - a) clear the device
  - b) set the device
  - c) always cause the output to change states
  - d) cause the output to assume a state dependent on the controlling (J-K or D) inputs.

6. A feature that distinguishes the J-K flip-flop from the D flip-flop is the a
- toggle condition
  - preset input
  - type of clock
  - clear input
7. A J-K flip-flop with  $J=1$  and  $K=1$  has a 10 kHz clock input. The Q output is d
- constantly HIGH
  - constantly LOW
  - a 10 kHz square wave
  - a 5 kHz square wave
8. Show the states of the 5-bit register in Figure below for the specified data input and clock waveforms. Assume that the register is initially cleared (all 0s). What will be the outputs ( $Q_4Q_3Q_2Q_1Q_0$ ) after 5 clocks?



9. Show the states of the 4-bit register (SRG 4) in Figure below for the specified data input and clock waveforms. Assume that the register initially contains all 1s. What will be the outputs ( $Q_3Q_2Q_1Q_0$ ) after 4 clocks?



10. With a 1 MHz clock frequency, eight bits can be parallel entered into a shift register **d**
- in  $8 \mu s$
  - in the propagation delay time of eight flip-flops
  - in  $1 \mu s$
  - in the propagation delay time of one flip-flop